

Unit 02.A Handout: Visualization

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Example 1: Histogram

The mile-run times of 50 students were recorded in minutes. Use the frequency table below to complete the relative-frequency column and draw a histogram.

4:55.21 Jake, 5:28.20 Alyssa, 7:15.03 Hannah, ..., 9:31.07 Tyler.

The class frequencies are given below:

Class	Frequency	Relative Frequency
[4, 5)	2	
[5, 6)	5	
[6, 7)	7	
[7, 8)	11	
[8, 9)	14	
[9, 10)	11	
Total	50	

Example 2: Dot Plot

The following data are breaking strengths, measured in gigapascals (GPa), for $n = 15$ carbon fibers. Construct a dot plot.

{0.1, 2.0, 2.3, 2.5, 3.2, 3.4, 3.4, 3.7, 3.9, 4.0, 4.1, 4.2, 4.2, 4.3, 4.4}

Example 3: Stem-and-Leaf Plot

(a)

Using the same data (Example 2), draw the stem-and-leaf plot.

{0.1, 2.0, 2.3, 2.5, 3.2, 3.4, 3.4, 3.7, 3.9, 4.0, 4.1, 4.2, 4.2, 4.3, 4.4}

(b)

Suppose, we have collected a random sample of 15 observations, 7 were from Method A and 8 were from Method B. Considering the following data, draw a side-by-side Stem-and-Leaf Plot. Describe the plots as right-skewed, left-skewed, or symmetric. Also, compare the mean and median of the observations of Method A and Method B.

Method A: $n = 7$	Method B: $n = 8$
$\{0.1, 2.0, 2.5, 3.2, 3.4, 3.4, 4.0\}$	$\{2.3, 3.7, 3.9, 4.1, 4.2, 4.2, 4.3, 4.4\}$

Example 4: Box Plot

Suppose, we have the following $n = 7$ observations:

$$\{0, 11, 13, 25, 34, 40, 81\}$$

For the above data set, find the values of the lower quartile, median, upper quartile, IQR, step, UIF, LIF, UOF, and LOF. Use this information to draw a boxplot.